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$$\frac{d}{dx}(x\sqrt{x})$$

$$f(x+h) = (x+h)\sqrt{x+h}$$

$$f(x+h)-f(x) = (x+h)\sqrt{x+h}-x\sqrt{x} = \frac{(x+h)^3-x^3}{(x+h)\sqrt{x+h}+x\sqrt{x}} = \frac{3x^2h+3xh^2+h^3}{(x+h)\sqrt{x+h}+x\sqrt{x}}$$

$$\frac{f(x+h)-f(x)}{h} = \frac{3x^2h+O(h^2)}{h((x+h)\sqrt{x+h}+x\sqrt{x})} = \frac{3x^2+O(h)}{(x+h)\sqrt{x+h}+x\sqrt{x}}$$

$$\lim_{h \rightarrow 0} \frac{3x^2+O(h)}{(x+h)\sqrt{x+h}+x\sqrt{x}} = \frac{3x^2}{2x^{\frac{3}{2}}} = \frac{3\sqrt{x}}{2}$$

References